

A
Micro Project Report
On

**“Implement and Compare Memory Allocation Strategies - First Fit,
Best Fit, Worst Fit”**

Submitted by
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Under the Guidance of
Prof. P. B. DATIR
In the Partial Fulfilment of Fifth Semester of
Diploma in Computer Engineering



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Affiliated to
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**Maharashtra State
Board of Technical Education**

Certificate

This is to certify that **Ms. Shaikh Farheen Hussain Mohammad** with **Roll No-16** has successfully completed Micro-project in course **Operating Systems (315319)** for the academic year 2025-26 as prescribed in the 'Micro-Project' during her tenure of completing Fifth Semester of Diploma Program in Computer Engineering from institute, Sandip Polytechnic with institute code 1167.

Place: Nashik

Enrollment No: 23611780178

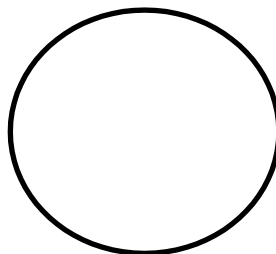
Date:

Exam Seat No:

Course Teacher

Head of the Department

Head of the Institute



ACKNOWLEDGEMENT

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Shaikh Farheen Hussain Mohammad (Roll No: 16)
Enrollment No 2:23611780186

Place: Nashik

Date:

Micro Project Report

1. Aim:

To study and compare different memory allocation strategies - First Fit, Best Fit, and Worst Fit, used by operating systems for efficient utilization of main memory through a diagrammatical representation.

2. Objectives / Benefits:

- To understand how the operating system allocates memory to processes.
- To compare the three memory allocation methods using diagrams.
- To study the effect of each strategy on fragmentation and space utilization.
- To develop visualization skills for memory management operations.

3. Introduction:

In a multiprogramming operating system, several processes reside in main memory at the same time. The operating system must allocate and deallocate memory efficiently to ensure smooth execution and minimal wastage of space.

Contiguous memory allocation is one of the simplest memory management methods, where each process occupies one continuous block of memory. The challenge is to decide which free memory block should be allocated to a new process.

Different allocation strategies make this decision in different ways. The three common strategies are:

1. **First Fit:** Allocates the first available block that is large enough.
2. **Best Fit:** Allocates the smallest available block that is sufficient for the process.
3. **Worst Fit:** Allocates the largest available block to the process.

Each of these techniques affects memory utilization and system performance differently.

4. Theory of Allocation Strategies:

(a) First Fit:

- The system scans memory from the beginning and allocates the first block that is large enough for the process.
- It is the simplest and fastest method.
- Can lead to moderate external fragmentation.

(b) Best Fit:

- The system searches all available blocks and chooses the smallest one that can fit the process.
- It minimizes unused space inside allocated blocks.
- It takes more time to search and may leave small fragments.

(c) Worst Fit:

- The system allocates the largest available block to the process.
- It tries to leave larger gaps available for future processes.
- It may lead to poor overall memory utilization.

5. Methodology:

To visualize how each strategy works, consider the following **memory map** and a new process **P4** with a **memory requirement of 6 KB**.

Given Memory Map:

O.S.
P1
<FREE> 12 KB
P2
<FREE> 19 KB
P3
<FREE> 7KB
Memory

(i) First Fit:

- Allocates the **first free block** that is large enough.
- The first available block of **12 KB** is selected.
- After allocation, **6 KB remains free**.

Result: Fast and simple allocation. Moderate fragmentation.

O. S.
P1
P4 6KB <FREE> 6KB
P2
<FREE> 19 KB
P3
<FREE> 7 KB

(ii) Best Fit:

- Allocates the **smallest free block** that can accommodate the process.
- The smallest sufficient block is **7 KB**.
- After allocation, **1 KB remains free**.

Result: Minimum unused space but slower allocation.

O. S.
P1
<FREE> 12 KB
P2
<FREE> 19 KB
P3
P4 6 KB <FREE> 1 KB

(iii) Worst Fit:

- Allocates the **largest free block** available.
- The largest block is **19 KB**.
- After allocation, **13 KB remains free**.

Result: May waste memory but leaves larger free areas

O. S.
P1
<FREE> 12 KB
P2
P4 6 KB <FREE> 13 KB
P3
<FREE> 7 KB

Observation Table:

Strategy	Free Block Selected	Free Space Left	Remark
First Fit	12 KB	6 KB	Fast allocation
Best Fit	7 KB	1 KB	Least waste, slower search
Worst Fit	19 KB	13 KB	Wastes large memory area

6. Advantages and Disadvantages:

(i) First Fit:

Advantages:

- Simple and fast.
- Minimum search time.

Disadvantages:

- Causes external fragmentation.

(ii) Best Fit:

Advantages:

- Reduces wasted space.
- Good for small processes.

Disadvantages:

- Slower search process.
- Creates many small unusable fragments.

(iii) Worst Fit:

Advantages:

- Leaves large unused spaces for future allocations.
- Useful for large process sizes.

Disadvantages:

- Poor memory utilization.
- May leave big blocks unused.

7. Comparative Analysis:

Strategy	Speed	Memory Utilization	Fragmentation	Efficiency
First Fit	Fast	Moderate	Moderate	Simple and quick
Best Fit	Slow	High	High internal	Space-efficient
Worst Fit	Moderate	Low	Low internal	Wastes space

8. Conclusion:

The study clearly highlights how memory allocation techniques differ in speed, efficiency, and memory utilization.

First Fit is best suited for systems that prioritize quick allocation and simplicity, as it requires the least searching time. However, it can lead to scattered unused memory blocks, known as fragmentation.

Best Fit performs better in terms of space utilization by minimizing unused memory inside allocated blocks, though it takes longer to search for the most appropriate space. It is more effective when memory optimization is more important than speed.

Worst Fit, on the other hand, allocates the largest available block, keeping larger gaps open for future processes. While this may initially appear efficient, it often leads to poor long-term memory utilization.

In summary, the choice of allocation method depends on the system's requirements. Real-time and general-purpose systems often use First Fit for faster execution, whereas resource-constrained environments prefer Best Fit for better memory conservation. Understanding these methods helps in designing an operating system that balances speed, efficiency, and resource management.

**Name & Signature of Course
Teacher**

(Prof. P. B. Datir)

Micro Project Evaluation Sheet

Name of Student: Shaikh Farheen Hussain Mohammad

Enrollment No: 23611780186

Name of Program: Computer

Semester: Fifth

Course Title: Entrepreneurship Development and Startups

Course Code: 315002

Title of Micro Project: Implement and Compare Memory Allocation Strategies - First Fit, Best Fit, Worst Fit

Course Outcomes Achieved:

Sr No	Characteristic to be Assessed	Poor Marks (1-6)	Average Mark (7-12)	Good Marks (13-19)	Excellent Marks (20-25)	Sub Total
(A) Process and Product Assessment (15 Marks)						
1	Relevance to the Course					
2	Literature Review/Information Collection					
3	Completion of the Targets per Project Proposal					
4	Analysis of Data & Representation					
5	Quality of Prototype/Model					
6	Report Preparation					
(B) Individual Presentation/Viva (10 Marks)						
7	Presentation					
8	Viva					

(A) Process and Product Assessment (15 Marks)	(B) Individual Presentation/Viva (10 Marks)	Total Marks (25 Marks)

Comments/Suggestions about Teamwork/Leadership/Interpersonal Communication (if any)

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**Name & Designation of Course
Teacher**
(Prof. P. B. Datir)